

TeachDB: A Knowledge Architecture for Human–AI Collaborative Thinking

Version 5.0 - Research Proposal

Human civilization has preserved information for thousands of years. But has humanity ever preserved reasoning itself? This proposal introduces TeachDB, a conceptual layer for AI architectures designed to treat reasoning—not information—as the fundamental unit of knowledge preservation. By enabling the continuous evolution of reasoning graphs, TeachDB aims to transition AI from transient conversational agents into persistent, co-evolving cognitive partners.

1. Introduction

Existing AI systems function primarily as retrieval engines or pattern predictors. They excel at accessing vast repositories of static information but struggle to capture the fluid, iterative process of human thought. TeachDB is not a database; it is a meta-architectural layer designed to externalize, structure, and evolve human reasoning.

2. Research Questions

- Can reasoning be represented and stored independently of natural language?
- Can human-AI dialogue continuously refine and expand formal reasoning graphs?

- Do persistent reasoning structures outperform document-based retrieval in complex decision-making?
- Can AI effectively preserve the evolution of human thought across years of interaction?

3. TeachDB System Architecture

The system operates as a recursive feedback loop:

Human → Dialogue → Reasoning Extraction Engine → Reasoning Graph → Knowledge Structure Generator → **TeachDB** → Knowledge Evolution Engine → LLM → Continuous Dialogue

The *Reasoning Extraction Engine* decomposes natural language into logical axioms. The *Knowledge Evolution Engine* manages the long-term consistency of the *Reasoning Graph*, allowing knowledge to adapt as new evidence emerges.

4. Reasoning Reconstruction: The Murakami Example

TeachDB does not simulate style; it reconstructs cognitive architecture. By analyzing the logic within his corpus, TeachDB builds a model of Murakami's thematic and structural reasoning. If prompted on AGI, the system applies this reconstructed logical framework to the new domain, providing an inference consistent with his underlying intellectual structure, rather than a mere stylistic mimicry.

5. Comparative Analysis

System	Unit	Function
RAG/GraphRAG	Documents/Entities	Information Retrieval

Vector DB	Embeddings	Semantic Matching
TeachDB	Reasoning Chains	Cognitive Synthesis

6. Potential Research Impact

TeachDB offers transformative potential across domains where reasoning provenance is critical: scientific discovery (mapping hypothesis evolution), medical diagnosis (tracking clinical reasoning), and corporate knowledge transfer (preserving institutional logic).

7. Limitations & Ethical Considerations

Key challenges include ensuring data provenance, mitigating bias in reasoning structures, and defining ownership over cognitive artifacts. Scalability remains an open problem in graph-based reasoning maintenance.